I developed multiple models to predict Customer Lifetime Value (LTV) using E_commerce data . Our baseline model, XGBoost without transformation, showed reasonable performance ($R^2 = 0.668$), but struggled with high outliers. After applying a log transformation to the target, the XGBoost model improved significantly ($R^2 = 0.778$, RMSE reduced by ~27%). A Random Forest model, while having a higher MAE, achieved the best R^2 (0.802), suggesting better variance capture. Based on use-case needs, XGBoost with log transformation is recommended for balanced prediction, while Random Forest is preferable for variance-sensitive applications.

🧾 Model Comparison: LTV Prediction

Model	MAE	RMSE	R^2 Score
XGBoost (No Log)	**318.75**	**4828.87**	**0.668**
XGBoost (Log Transformed)	**284.56**	**3544.34**	**0.778**
Random Forest	**403.26**	**3736.42**	**0.802**

🔍 Metric Explanation

📉 1. MAE – Mean Absolute Error

🔍 What it is:

MAE measures the **average absolute difference** between the actual values and the predicted values.

📐 Formula:

$$\text{MAE} = \frac{\sum_{i=1}^{n} |y_i - x_i|}{n}$$

- yi= actual value
- Xi = predicted value
- n = number of predictions

✅ Why it's useful:

- Easy to understand: it tells you how wrong your model is, on average.
- All errors are weighted equally.
- Units are the **same as the target variable** (e.g., dollars).

⚠️ **Limitations:**

- Doesn’t penalize large errors heavily. A huge mistake counts the same as a small one.

📉 2. RMSE – Root Mean Squared Error

🔍 **What it is:**

RMSE also measures the difference between predicted and actual values, **but squares the errors** before averaging and then takes the square root.

📐 **Formula:**
RMSE=

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N} (Predicted_i - Actual_i)^2}{N}}$$

✅ **Why it’s useful:**

- Penalizes **large errors more heavily**, so it’s sensitive to outliers.
- Also in the same units as the target variable.

⚠️ **Limitations:**

- Can exaggerate the effect of a few bad predictions.

📊 3. R^2 Score – Coefficient of Determination

🔍 **What it is:**

R^2 tells you how much of the **variation** in the target variable your model explains.

📐 **Formula:**

$$R^2 = 1 - \frac{\sum_{i=1}^{m} (X_i - Y_i)^2}{\sum_{i=1}^{m} (\bar{Y} - Y_i)^2}$$

- The denominator is the **total variance** in actual values.
- The numerator is the **residual variance** (how much error is left in the model).

✅ **Why it's useful:**

- Gives a **single metric between 0 and 1**:
  - 0 → model predicts no better than the average
  - 1 → perfect prediction
- Helps judge **overall model performance**.

⚠️ **Limitations:**

- Can be misleading with non-linear models.
- Can be negative if the model performs worse than predicting the mean.

📘 Summary Table

Metric	Interpretation	Good Value	Notes
MAE	Avg error (absolute)	Lower is better	Easy to interpret
RMSE	Penalizes big errors	Lower is better	Sensitive to outliers
R^2 Score	Explained variance	Closer to 1	Overall fit of the model

📊 Model Analysis & Interpretation

✅ 1. XGBoost (Without Log Transformation)

- **MAE: 318.75**, **RMSE: 4828.87**, **R^2: 0.668**
- This is your base model. It does a decent job predicting LTV.
- However, the **RMSE is quite high**, suggesting it **struggles with large outliers** - big spenders are not predicted well.
- The R^2 of 0.668 means the model explains **66.8% of the variation** in customer LTV.

Why?

- XGBoost is sensitive to skewed data.
- LTV has natural outliers (some customers spend much more).
- Without log transformation, the model is “pulled” by these large values, making overall predictions less stable.

✅ 2. XGBoost (Log-Transformed Target)

- **MAE: 284.56**, **RMSE: 3544.34**, **R^2: 0.778**
- Applying a **log transformation to the target (Monetary)** improves the model significantly.
- Lower MAE and RMSE, and R^2 jumps to 77.8%.
- This model **reduces the influence of high-value outliers**, helping it generalize better.

Why it works:

- The log transform **compresses the scale** of high values.
- It stabilizes variance and makes the LTV distribution more normal, which XGBoost handles better.
- After predicting, you **`expm1()`** to bring the values back to real-world currency.

✅ 3. Random Forest (No Log)

- **MAE: 403.26**, **RMSE: 3736.42**, **R^2: 0.802**
- MAE is higher than both XGBoost models, but **R^2 is the highest** - meaning it captures **more variance** in the LTV.
- Random Forest is **robust to outliers**, which is why it performs well even without log transformation.

Key Insight:

- While Random Forest overpredicts or underpredicts slightly more often (higher MAE), it **gets the overall trend right better** (higher R^2).
- It may be a better choice **if interpretability and stable performance across all segments is your priority**.

📈 Final Recommendation

- If you want **balanced accuracy with interpretability**, go with **XGBoost with Log Transformation**. It performs well on both MAE and RMSE and avoids extreme prediction errors.
- If your goal is **capturing as much variance as possible**, or you're building a **ranking system** (e.g., top LTV customers), **Random Forest** is very strong despite its higher MAE.
- Avoid **plain XGBoost without log transform** unless you further tune it or handle outliers separately.